

For reprint orders, please contact: reprints@futuremedicine.com

Comprehensive pediatric care of rare bleeding disorders

Muriel Giansily-Blaizot[†] & Jean-François Schved

Although less common than hemophilia or von Willebrand disease, inherited rare bleeding disorders, comprising afibrinogenemia and deficiencies in factors II, V, VII, X, XI, XIII or V plus VIII combined, or in vitamin K-dependent coagulation factors, may lead to severe bleeding episodes such as recurrent hemarthroses and neonatal intracranial or gastrointestinal hemorrhage. Consanguinity significantly increases the risk of the occurrence of all rare bleeding disorders that are associated with an autosomal recessive pattern of inheritance. Each of the disorders is characterized by a wide interindividual variation in clinical phenotype and a large mutational spectrum with no clear correlation between the phenotype and genotype. Replacement therapy relies on specific molecules or concentrates (afibrinogenemia, factor VII, XI and XIII deficiencies), on a mixture of different concentrates that are otherwise known as a prothrombin complex, which contains factors II, VII, IX and X, or on fresh frozen plasma. International consensus guidelines for treatment modalities are progressing; however, guidelines for prophylaxis, especially in pediatric patients, are lacking.

Inherited rare bleeding disorders (RBD), including afibrinogenemia and deficiencies in factor (F) II, V, VII, X, XI, XIII or V plus VIII combined, or in vitamin K-dependent coagulation factor (VKDC), generally represent 5% of all inherited coagulation disorders [1] with slight variations occurring between countries depending, in particular, on the prevalence of consanguinity in different racial groups. In that way, RBDs differ from hemophilia A and B where the prevalence appears to be similar in all races. The prevalence of RBDs ranges from one in 3,000,000 individuals for the rarest (VKDC or FXIII deficiency) to one in 300,000 individuals for the most common (FVII or FXI deficiencies) homozygous or compound heterozygous forms. All of the above disorders are transmitted in an autosomal recessive manner, implying that the heterozygous carriers are asymptomatic. This is true for the combined FV plus FVIII and VKDC defects where heterozygotes have normal clotting factor levels. Contrary to this, heterozygous carriers for the single deficiency RBDs generally display reduced levels of their respective factor. Those individuals are usually asymptomatic, but symptomatic heterozygous carriers have been widely reported with the FXI deficiency [2] and mild bleeding features might also be observed in FII-, FVII- or FX-deficient heterozygotes [3–5]. Nevertheless, the autosomal recessive trait implies that the risk of presenting with a RBD is increased where consanguineous marriages are common, explaining the aforementioned variable incidences between

countries. In addition, there may be no preceding family history to suggest the diagnosis and an affected child may be part of a kindred with very few affected individuals – heterozygous patients usually being asymptomatic. As bleeding symptoms in children manifest in a manner that is distinct to those in adults, pediatricians and neonatologists need to be aware of these differences and thoroughly investigate every unexpected bleeding event that presents early in life, since it may potentially represent early markers of severe inherited bleeding disorders. Indeed, the most severely affected patients often present with life-threatening hemorrhages within a few days of birth, including intracranial hemorrhages (ICHs) or gastrointestinal hemorrhages. Inadequate or delayed treatment of such neonatal bleeding may increase the risk of mortality or significant long-term disability. Later in life, recurrent joint bleeding may lead to the development of chronic arthropathy. Emphasis should be placed on early diagnosis, replacement treatment and prophylaxis. Two entire issues of the journals *Haemophilia* (2008) and *Seminars in Thrombosis and Hemostasis* (2009) are devoted to the RBDs and are recommended for supplementary information.

Clinical phenotype: can each RBD be characterized in terms of typical bleeding features?

Patients affected by RBDs present with a wide range of clinical phenotypes – being asymptomatic at one end of the spectrum and suffering fatal

[†]Author for correspondence:
Laboratoire d'hématologie,
CHU de Montpellier,
80 Avenue Augustin Fliche,
34095 Montpellier Cedex 5,
France
Tel.: +33 467 337 033
Fax: +33 467 337 036
m-giansily@chu-montpellier.fr

Keywords

- afibrinogenemia • FII deficiency
- FVII deficiency • FX deficiency
- FXI deficiency • FXIII deficiency
- intracranial hemorrhage
- pediatrics • rare bleeding disorder

future
medicine part of fsg

cerebral nervous system bleeding at the other. As discussed above, heterozygous carriers of RBDs are usually asymptomatic or might present with mild bleeding manifestations. Therefore, the occurrence of severe spontaneous bleedings in RBD heterozygotes should prompt clinicians to rule out an additional hemostasis disorder. On the other hand, children homozygous for a RBD can be either asymptomatic or exhibit mild, moderate or severe bleeding. The bleeding tendency can first appear at any age, although the more severely affected patients may present with life-threatening hemorrhages within the first few days of birth.

Even when focusing on the severe forms of RBDs, no one bleeding feature definitively characterizes any one of the RBDs discussed in this review, although some may be more suggestive of a specific factor deficiency than others. First, nervous system bleeding occurring within the first few months of birth is more frequently associated with severe FII, FVII, FX and FXIII deficiencies than with severe FV deficiency or afibrinogenemia [6]. Second, delayed umbilical stump bleeding is suggestive of FXIII deficiency (occurring in 80% of cases) [7] or afibrinogenemia [8]. Finally, hemarthrosis more frequently occurs in association with FV deficiency than with any other RBD. Among this group of clotting disorders, inherited FXI deficiency stands apart as it is most commonly an injury-related bleeding disorder [2]. The main characteristics of each RBD are summarized in TABLE 1.

Severe clinical features including spontaneous ICH, whilst being rare, do account for the mortality associated with this group of diseases. The incidence is estimated at 34, 21 and 7% in FXIII [9], FX [5] and FVII [10] deficiencies, respectively, with a high prevalence of consanguinity among the parents of affected newborns. Clinical presentation varies considerably between published case reports, irrespective of the deficient clotting factor. ICH can represent the first symptom, and can be observed as early as in the first few days following birth [11] or later on in infancy after a normal prenatal period and delivery [12]. However, signs of a congenital bleeding disorder, such as bleeding mucosae including epistaxis of varying degrees of severity and easy bruising, or more suggestively, soft tissue hematomas or umbilical stump bleeding, often occur in the weeks preceding a cerebral bleed [13,14]. Therefore, the presence of unexpected bleeding symptoms, especially if the parents are consanguineous, should prompt physicians to perform clotting tests. Equally important is the diagnosis

of a congenital bleeding disorder, which should be considered in children with spontaneous ICHs, even in those who had no prior episode of extracerebral hemorrhage [12].

As mentioned previously, bleeding histories of children demonstrate marked differences to those of adults. For example, postsurgical bleeding, a classical symptom that is generally considered to be of value to bleeding histories, is not prevalent in pediatric patients since patients are often too young to have suffered any hemostatic challenge. Therefore, in pediatric cases, particular attention must be given to any provoked bleeding, for example, at the time of circumcision or umbilical cord bleeding.

Biological phenotype: which clotting tests should be conducted for a quick diagnosis?

In severe forms of RBDs, early diagnosis is crucial due to the risks associated with ICH. Since both age and sex have a profound influence on the hemostatic cascade, data from the results of the same tests conducted in age- and sex-matched healthy subjects should be made more easily accessible in order to accurately interpret the clotting test results. The hemostatic system matures during the first weeks and months following birth, with most hemostatic parameters reaching adult values by 6 months of age (meticulously reviewed in [15]). In the neonatal period, only RBD homozygotes can be diagnosed, whereas levels of the missing factor in heterozygotes may overlap with the normal range [15]. The first-line coagulation tests in common use are tests for the activated partial thromboplastin time, prothrombin time and thrombin clotting time or the fibrinogen measurement. Prolongation of these clotting tests, when they are corrected by the addition of an equal volume of normal plasma to that of the test plasma, is indicative of either a single factor or multiple factor deficiency [15]. The tests generally allow the screening of all RBDs except FXIII deficiency (TABLE 1). Severe FXIII (fibrin-stabilizing factor) deficiency differs as it leads to impaired fibrin cross-linking and the production of defective and unstable blood clots that do not affect the clotting end point of routine coagulation tests. Therefore, specific screening tests, including the urea clot solubility test or, even more easy to conduct, specific FXIII activity assays, should be performed in patients who are suspected of having FXIII deficiency [7].

The vast majority of bleeding problems encountered in the neonatal period are due to acquired hemostatic defects including vitamin K

Table 1. Phenotypic and molecular characteristics and treatment modalities of each rare bleeding disorder.

Factor deficiency	Clinical features	PT	APTT	TCT	Mutational spectrum	Treatment	Ref.
Afibrinogenemia	Large heterogeneity Usual onset at neonatal period: bleedings of the umbilical cord, GI bleedings and ICHs Onset at a later age is possible (beware of spleen ruptures)	↑	↑	↑	Higher incidence of mutations in the <i>FGA</i> gene (compared with the <i>FGB</i> and <i>FGG</i> genes) There are two common mutations in Europeans in <i>FGA</i> : c.510 + 1G>T and 11-kb deletion	Fibrinogen factor concentrates either on-demand or in prophylaxis in cases of neonatal bleeding events	[8]
Hypo-fibrinogenemia	Usually asymptomatic	Fibrinogen levels of approximately 1.0 g/l TCT is the most sensitive assay			Heterozygotes for afibrinogenemia mutations	Hypodysfibrinogenemia patients may sometimes need replacement therapy during surgery	[8]
FII	FII:C <20%: spontaneous hemorrhage (ICHs and hemarthrosis) FII:C >20%: heterogeneous and moderate bleeding	↑	↑	N	Large heterogeneity [†]	PCC or FFP	[3–6]
FV	FV:C <20%: oral cavity bleedings (60%), epistaxis hematomas and hemarthroses (25%); ICHs and GI bleeding events are extremely rare FV:C >20%: heterogeneous, mucous membrane bleeding or asymptomatic	↑	↑	N	Up to 50 FV mutations have been reported (Tyr1702Cys is recurrent in Italy) [‡]	FFP or platelets	[29]
FV plus VIII	Provoked skin and mucous membrane bleeding events	↑	↑	N	Mutations have been found in <i>LMAN 1</i> in two thirds of mutations Mutations have been found in <i>MCFD 2</i> in a third of mutations	On-demand treatment with sources of both FV and FVIII In the absence of recommendations, hemostatic levels of 30 and 20% should be used for FVIII and FV, respectively	[16]

Percentage values in the 'Clinical features' column refer to the prevalence of the bleeds.

[†]See the Coagulation Serine Protease Mutation Database [33].

[‡]Mutation database available on request from h.i.vos@lumc.nl.

[§]Mutation database available on request from m-giansily@chu-montpellier.fr.

[¶]The Factor XI Deficiency Mutation Database is available at [102].

^{**}See the FXIII Registry Database [103].

^{††}: Prolongation of the corresponding clotting test. APTT: Activated partial thromboplastin time; C: Coagulant activity; F: Factor; FFP: Fresh frozen plasma; GGC: γ-glutamyl carboxylase; GI: Gastrointestinal;

ICH: Intracranial hemorrhage; LMAN 1: Leguminous mannose-binding lectin gene; MCED 2: Multiple coagulation factor deficiency gene 2; N: Clotting test within normal ranges; PCC: Prothrombin complex concentrate;

PI: Prothrombin time; TCT: Thrombin clotting time; VKORC1: Vitamin K epoxide reductase unit 1.

Table 1. Phenotypic and molecular characteristics and treatment modalities of each rare bleeding disorder (cont.).

Factor deficiency	Clinical features	PT	APTT	TCT	Mutational spectrum	Treatment	Ref.
FVII	Wide heterogeneity with a poor relationship between FVII:C levels and bleeding tendency ICHs (7%), hemarthroses (22%) and epistaxis (83%)	↑	N	N	Large heterogeneity, up to 250 mutations reported [§]	Treatment is on-demand or prophylactic for early diagnosed patients and those with severe bleeding (newborns and children) using recombinant activated FVII or FVII concentrates when available; alternative therapy includes PCCs	[30]
FX	The most severe rare bleeding disorder FX:C <2%: life-threatening CNS or GI hemorrhage 2% < FX:C < 15%: severe bleeding events, hemarthrosis and hematoma FX:C >20%: asymptomatic (70% of all cases) and provoked bleeding	↑	↑	N	Large heterogeneity (Gly[-20]Arg and Gly380Arg are recurrent in Latin America) [†]	Freeze-dried FIX and FX concentrates (in some countries) or PCC or FFP	[5,23]
FXI	Injury-related bleeding events Increased risk when local fibrinolytic potential is high (urogenital tract and oral cavity)	N	↑	N	See the Factor XI Deficiency Mutation Database [†]	FXI concentrates	[2,31]
FXIII	Umbilical stump bleeding (80%), ICHs (30%), hematomas (27%) and impaired wound healing (14–29%)	N	N	N	Mutations have been found in both <i>FXIII-A</i> and <i>FXIII-B</i> genes [#]	FXIII concentrate	[7]
VKDC	Large heterogeneity ranging from fatal ICHs at birth to provoked hemorrhagic manifestations in childhood associated with antibiotic therapy leading to decreased gut vitamin K production	↑	↑	N	Mutations have been found in both <i>VKORC1</i> and <i>GGCX</i> genes Diagnosis can be delayed owing to the routine administration of vitamin K1 to newborns	Large heterogeneity in the responsiveness to oral or even parenteral vitamin K1 Where ineffective, factor replacement is mandatory (PCC or FFP)	[32]

Percentage values in the 'Clinical features' column refer to the prevalence of the bleeds.

[†]See the Coagulation Serine Protease Mutation Database [33].

[‡]Mutation database available on request from h.i.vos@lumc.nl.

[§]Mutation database available on request from m-giansily@chu-montpellier.fr.

[#]The Factor XI Deficiency Mutation Database is available at [102].

^{*}See the FXIII Registry Database [103].

†: Prolongation of the corresponding clotting test. APTT: Activated partial thromboplastin time; C: Coagulant activity; F: Factor; FFP: Fresh frozen plasma; GGCX: γ-glutamyl carboxylase; GI: Gastrointestinal; ICH: Intracranial hemorrhage; LMAN 1: Leguminous mannose-binding lectin gene; MCFD 2: Multiple coagulation factor deficiency gene 2; N: Clotting test within normal ranges; PCC: Prothrombin complex concentrate; PT: Prothrombin time; TCT: Thrombin clotting time; VKORC1: Vitamin K epoxide reductase unit 1.

deficiency or disseminated intravascular coagulation [15]. Inherited bleeding disorders are rare. Acquired clotting factor deficiencies owing to the presence of inhibitors are unlikely to occur in children. As soon as a rare inherited bleeding disorder is suspected, specific factor assays should be performed to identify the corresponding RBD. From a clinical stand point, the factor activity level is sufficient to make a correct diagnosis. For a more detailed biological phenotype, the factor antigen level may be assessed, thus allowing the classification of the deficiency. Type I is usually caused by defects in the expression and/or secretion of the protein, and in type II, the factor is present but not functioning. In type I deficiencies, factor coagulant activity and antigen levels decrease in parallel, whereas in type II deficiencies, there is a discrepancy between the reduced factor coagulant activity levels and up to normal antigen levels. A more detailed protein characterization may require a full spectrum of tests to be conducted, which can only be performed in specialized laboratories.

Certain phenotypic characteristics are required to be mentioned that may require further investigation. First, levels of FVIII should be tested in individuals with reduced FV levels to exclude the possibility of combined FV plus FVIII deficiency. In these cases, FV and FVIII levels are usually within the mild ranges although levels below 10% have been reported [16]. Second, some FVII variants (R304Q, FVII Padua, R79Q and G331D) display different FVII coagulant activities depending on the species-derived thromboplastin preparation used. The most common example concerns FVII Padua (R304Q), which frequently occurs in southern Europe and Africa. When present in the homozygous state, this variant may exhibit activity levels as low as 2% when rabbit tissue factor is used, whereas levels can reach 30–35% with recombinant human tissue factor [17]. Therefore, recombinant human tissue factor preparations and highly sensitive clotting assays are recommended for the correct assignment of FVII:coagulant activity (compared with the levels of antigen) in these cases [18].

Genotype: is molecular diagnosis useful in common practice?

As discussed previously, each RBD has an autosomal recessive pattern of inheritance, except for the FXI deficiency, which can mimic a dominant trait since bleeding also occurs in heterozygous individuals. The molecular diagnosis of all RBDs, apart from the FV plus FVIII and VKDC combined deficiencies, is based on the

identification of the mutation located within the gene encoding the respective missing factor. The FV plus FVIII deficiency is caused by mutations in two different genes encoding proteins involved in the transportation of both FV and FVIII, namely the leguminous mannose-binding lectin gene (*LMAN 1*) and the multiple coagulation factor deficiency gene 2 (*MCFD 2*) [16]. VKDC deficiency is caused by mutations located in the genes encoding proteins involved in vitamin K metabolism, namely vitamin K epoxide reductase unit 1 (*VKORC1*) and γ -glutamyl carboxylase (*GGCX*) [19,20]. Databases available online now provide lists of causative mutations for almost every RBD (TABLE 1).

The considerable interindividual heterogeneity in bleeding phenotypes is the result of a combination of parameters, including the molecular heterogeneity of the RBD but also other factors, both genetic and environmental. Indeed, no clear genotype–phenotype relationship has emerged between the clinical expression of the deficiency, plasma activity levels of the corresponding missing factor and the respective mutations, whichever RBD is considered. Furthermore, the severity of bleeding can be highly variable, even among patients bearing the same genotype. Therefore, one could consider that the main interest of molecular diagnosis is in the scientific community. Exhaustive genotyping increases the amount of available data on each RBD, which in turn helps to characterize the mutational spectrum, highlighting recurrent mutations within specific geographical areas, and this improves our chances of discovering genotype–phenotype correlations. However, that is partly true, especially for pauci- or asymptomatic patients, but it must be kept in mind that early molecular diagnosis can be useful for severely affected children. When a child manifests severe bleedings of any kind (ICHs, gastrointestinal bleedings or recurrent hemarthroses), they must be considered at risk for further life-threatening bleedings and therefore, prophylactic replacement therapies must be initiated. It is more complicated when a severe RBD is diagnosed in the early days of life owing to ‘mild’ bleeding features. In those cases, the option of long-term prophylaxis could be supported by molecular biology. Indeed, null mutations, when combined in the homozygous or compound heterozygous state, have been demonstrated, particularly in FVII deficiency, to be associated with a high risk of developing cerebral bleeding within the first few days or weeks of life (reviewed in [21]). The deleterious mutations involved include nonsense

mutations, splice-site mutations located in the invariant AG or GT dinucleotides, complete or partial gene deletions and frame-shift or missense mutations occurring on critical residues. In these cases, early identification of the underlying null mutation could be of great interest to individuals in whom a RBD manifests early in life, as this will improve their risk assessment [22]. Therefore, international molecular diagnosis networks should be developed, as DNA samples can easily be transferred to a reference center or research team.

Elsewhere, molecular diagnosis is essential for genetic counseling and prenatal diagnosis. Prenatal diagnosis in the event of a subsequent pregnancy is generally only proposed to families with an affected child who suffers from very severe manifestations, for example, intracerebral hemorrhage or recurrent hemarthroses. Prenatal diagnosis is sometimes considered when both parents are heterozygous carriers of a null mutation where there is a high risk of having a severely affected child [23]. The prenatal diagnosis of several RBDs has already been performed [24,25].

Treatment: which drug & which dose?

Treatments should be divided into two groups: replacement therapy and nonspecific drugs. For all of the disorders studied, nonspecific antihemorrhagic drugs can be helpful for preventing hemorrhages or decreasing their frequency. The most widely used antihemorrhagic drug, which is particularly used in cases of FXI deficiency, is tranexamic acid. However, replacement therapies are required in all cases involving hemorrhagic complications. These involve the use of either a specific concentrate (FXI, FXIII and fibrinogen), when available, or a mixture of different concentrates (either prothrombin complex concentrate [PCC] containing FX, FII, FIX and FVII, or fresh frozen plasma [FFP]). Thus, the treatment requires blood-derived products. Therefore, while viral and pathogen inactivation dramatically reduces the risk of transfusion-transmitted disease, vaccination, when available, is recommended, notably against the hepatitis viruses. The availability of treatment differs according to geographical location; however, the possible choices are as follows:

- Fibrinogen deficiency: fibrinogen concentrate, cryoprecipitate or FFP;
- FII deficiency: PCC or FFP;
- FV deficiency: FFP or platelets;

- FVII deficiency: recombinant factor VIIa (rFVIIa) and FVII concentrates can be used when available. PCC can be used as an alternative therapy;
- FX deficiency: freeze-dried human coagulation FIX and FX concentrate is available in some countries. PCC or FFP can also be used;
- FXI deficiency: use FXI concentrates. When FXI concentrates are not available, FFP can also be used;
- FXIII deficiency: FXIII concentrate.

There is no standard dose for any of these drugs – the dose infused must be calculated according to the basal level and the level expected after infusion. As such, a pharmacokinetic study in the patient is sometimes required to determine recovery and drug half-life. It should be noted that the expected postinfusion level is not the same for all drugs.

Treatment: when?

There is no international consensus concerning treatment modalities for RBDs. Severe hemorrhagic complications require immediate infusion of replacement therapy or an equivalent. The desired plasma level is likely to differ depending on which factor is deficient, and yet, except for FXIII and fibrinogen, a postinfusion level superior to 30% is recommended. For FXIII, a 5% plasma level is probably enough, while for fibrinogen, the level of security ranges between 0.8 and 1 g/l.

Surgery is sometimes necessary in patients with a RBD. Preoperative screening constitutes the main circumstance of diagnosis. However, the plasma level of each factor that would ensure a safe surgical procedure is poorly documented. There is a general trend of proposing replacement therapy in conjunction with all major surgical procedures, whatever the basal factor level. Yet reports of patients having previously undergone surgery without replacement therapy, when the diagnosis of factor deficiency had not been made, are frequent [21]. Many mildly affected patients only need tranexamic acid for minor-to-moderate surgeries [2]. With this in mind, most patients with a mild deficiency may not require replacement therapy to undergo surgery. The problem lies in determining the minimum plasma level required to ensure that surgery is safe without substitution. This level should vary depending on the form of deficiency and probably varies between patients. However, in every case, all invasive procedures should take

place at a specialized center and be carried out by physicians who are experienced in operating on patients with hemophilia.

Another critical question concerns the scheme of long-term treatment and prophylactic measures. Most of the knowledge of congenital hemorrhagic disease management comes from hemophilia, for which there is a large consensus to recommend prophylaxis as a reference treatment rather than an on-demand therapy. In hemophilia, prophylaxis consists of infusions of factor concentrates two- to three-times a week, or on alternate days, even in the absence of bleeding, with the aim of obtaining minimal residual plasma levels of the factor in question in order to prevent hemorrhage. A randomized study confirmed the superiority of this therapeutic approach for preventing chronic arthropathy in hemophilia A patients [26]. In RBDs, prophylaxis therefore likely represents the most suitable therapeutic approach; however, the exact protocol in terms of the dose and frequency of infusions now needs to be determined. For instance, prophylaxis in FXIII deficiency is recommended once a month [7], whereas in FVII deficiency, two- or three-times a week appears to be a good compromise, despite the short half-life of rFVIIa [27].

Finally, all the therapeutic decisions concerning this group of patients should be made by a team of specialists, thus requiring the follow-up of all affected children in a specialized reference center for the treatment of hemophilia.

Future perspective

Future perspectives should focus on three major issues: improving the early diagnosis of RBDs, searching for clinical or biological markers that are capable of predicting any bleeding risk and developing specific therapies where they are not currently available. First, pediatric-specific bleeding scores could be of great interest for both the screening of RBDs and the evaluation of bleeding risk prior to surgery. Many efforts have been made to develop quantitative scoring systems; these systems have recently undergone pediatric-specific modifications and are currently being evaluated in large multicentric studies (reviewed in [28]). However, bleeding histories are subjective and sometimes difficult to obtain since a child may not have undergone any hemostatic challenge at the time of examination. Therefore, other biological parameters such as the thrombin generation time should also be tested to predict the bleeding risk in children. Second, while improving the treatment of RBDs represents a difficult challenge, urgent efforts should be focused on developing a concentrate to treat patients who are affected by FV deficiency or the combined FV plus FVIII deficiencies. Currently, no FV concentrate exists and FV is only available through FFP. However, unfortunately, numerous countries do not have access to virus-inactivated FFP. Finally, the major issue surrounding RBDs is the lack of information available in the literature, with data being from short series

Executive summary

Clinical phenotype: are any of the rare bleeding disorders associated with typical bleeding features?

- No one bleeding feature is characteristic of any of the rare bleeding disorders.
- Spontaneous intracranial hemorrhages, although rare, account for the mortality.
- The risk of the rare yet severe coagulation defects increases in offspring of consanguineous parents.

Biological phenotype: which clotting tests should be conducted for a quick diagnosis?

- Every unexpected bleeding event presenting in early infancy must be investigated urgently using first-line coagulation tests, investigating the activated partial thromboplastin time, prothrombin time and thrombin clotting time.
- Specific screening tests, including the urea clot solubility test or specific FXIII activity assay, should be performed for patients who are suspected of having a FXIII deficiency.

Genotype: is molecular diagnosis useful for common practice?

- Molecular diagnosis is essential for genetic counseling and can be helpful for the most severe forms of rare bleeding disorders, including intracranial hemorrhage.

Treatment: which drug, what dose and when?

- Severe hemorrhagic complications require immediate infusion of replacement therapy or equivalent.
- All invasive procedures should be performed by surgeons with experience in operating on hemophilic patients within a specialized center.

Future perspective

- Pediatric-specific bleeding scores and biological screening tests that are able to predict the bleeding risk now need developing.
- International registries, including data from both adults and children, are of great interest with such rare disorders.

or isolated clinical cases. The constitution of international molecular and therapeutic registries, such as the International Rare Bleeding Disorders Database [101], should provide better access to information regarding the frequency of clinical manifestations and the relationships between genotypes and phenotypes, and may eventually help to regroup experts in the field with a view to forming consensual therapeutic measures.

Bibliography

Papers of special note have been highlighted as:

• of interest

•• of considerable interest

- Mannucci PM, Duga S, Peyvandi F: Recessively inherited coagulation disorders. *Blood* 104, 1243–1252 (2004).
- **Provides an extensive review of the rare bleeding disorders in both adults and children.**
- Salomon O, Steinberg DM, Seligshon U: Variable bleeding manifestations characterize different types of surgery in patients with severe factor XI deficiency enabling parsimonious use of replacement therapy. *Haemophilia* 12, 490–493 (2006).
- Lancellotti S, De Cristofaro R: Congenital prothrombin deficiency. *Semin. Thromb. Hemost.* 35, 367–381 (2009).
- Herrmann FH, Wulff K, Auerswal DG: Factor VII deficiency: clinical manifestation of 717 subjects from Europe and Latin America with mutations in the factor 7 gene. *Haemophilia* 15, 267–280 (2009).
- Herrmann FH, Auerswald G, Ruiz-Saez A *et al.*: Greifswald Factor X Deficiency Study Group: Factor X deficiency: clinical manifestation of 102 subjects from Europe and Latin America with mutations in the factor 10 gene. *Haemophilia* 12, 479–89 (2006).
- Acharya SS, Coughlin A, Dimichele DM; North American Rare Bleeding Disorder Study Group: Rare Bleeding Disorder Registry: deficiencies of factors II, V, VII, X, XIII, fibrinogen and dysfibrinogenemias. *J. Thromb. Haemost.* 2, 248–56 (2004).
- Karimi M, Berczky Z, Cohan N, Muszbek L: Factor XIII deficiency. *Semin. Thromb. Hemost.* 35, 426–438 (2009).
- de Moerloose P, Neerman-Arbez M: Congenital fibrinogen disorders. *Semin. Thromb. Hemost.* 35, 356–366 (2009).
- **Highly informative review on the various forms of inherited fibrinogen disorders.**
- Ivaskevicius V, Seitz R, Kohler HP *et al.*: International registry on factor XIII deficiency: a basis formed mostly on European data. *Thromb. Haemost.* 97, 914–921 (2007).
- Mariani G, Herrmann FH, Dolce A *et al.*: International Factor VII Deficiency Study Group: Clinical phenotypes and factor VII genotype in congenital factor VII deficiency. *Thromb. Haemost.* 93, 481–487 (2005).
- Landau D, Rosenberg N, Zivelin A *et al.*: Familial factor VII deficiency with foetal and neonatal fatal cerebral haemorrhage associated with homozygosity to Gly180Arg mutation. *Haemophilia* 15, 774–778 (2009).
- Thachil JV, Caswell M, Keenan R *et al.*: Factor X deficiency presenting as a pseudotumor. Case report. *J. Neurosurg.* 104(3 Suppl.), 202–205 (2006).
- Giansily-Blaizot M, Aguilar-Martinez P, Briquel ME *et al.*: Two novel cases of cerebral haemorrhages at the neonatal period associated with inherited factor VII deficiency, one of them revealing a new nonsense mutation (Ser52Stop). *Blood Coagul. Fibrinolysis* 14, 217–220 (2003).
- Khalife H, Muwakkit S, Al-Moussawi H *et al.*: Spontaneous splenic rupture in a patient with factor XIII deficiency and a novel mutation. *Pediatr. Blood Cancer* 50, 113–114 (2008).
- Williams MD, Chalmers EA, Gibson BE *et al.*: The investigation and management of neonatal haemostasis and thrombosis. *Br. J. Haematol.* 119, 295–309 (2002).
- **Establishes guidelines for clotting tests of neonates, both premature and full-term, performed up to 6 months of age.**
- Zhang B, Spreafico M, Zheng C *et al.*: Genotype-phenotype correlation in combined deficiency of factor V and factor VIII. *Blood* 111, 5592–600 (2008).
- Girolami A, Fabris F, Dal Bo Zanon R, Ghiotto G, Burul A: Factor VII Padua: a congenital coagulation disorder due to an abnormal factor VII with a peculiar activation pattern. *J. Lab. Clin. Med.* 91, 387–395 (1978).
- Giansily-Blaizot M, Verdier R, Biron-Adréani C *et al.*: Analysis of biological phenotypes from 42 patients with inherited factor VII deficiency: can biological tests predict the bleeding risk? *Haematologica* 89, 704–709 (2004).
- Rost S, Fregin A, Ivaskevicius V *et al.*: Mutations in *VKORC1* cause warfarin resistance and multiple coagulation factor deficiency type 2. *Nature* 427, 537–541 (2004).
- Darghouth D, Hallgren KW, Shtofman RL *et al.*: Compound heterozygosity of novel missense mutations in the γ -glutamyl-carboxylase gene causes hereditary combined vitamin K-dependent coagulation factor deficiency. *Blood* 108, 1925–1931 (2006).
- Giansily-Blaizot M, Schved JF: Potential predictors of bleeding risk in inherited factor VII deficiency. Clinical, biological and molecular criteria. *Thromb. Haemost.* 94, 901–906 (2005).
- Hennewig U, Eisert S, Wulff K *et al.*: Long-term FVII substitution in a preterm infant with severe gastrointestinal bleeding and FVII deficiency due to a homozygous donor splice mutation IVS4+1G→A. *Pediatr. Hematol. Oncol.* 23, 129–133 (2006).
- Menegatti M, Peyvandi F: Factor X deficiency. *Semin. Thromb. Hemost.* 35, 407–415 (2009).
- Neerman-Arbez M, Vu D, Abu-Libdeh B *et al.*: Prenatal diagnosis for congenital afibrinogenemia caused by a novel nonsense mutation in the *FGB* gene in a Palestinian family. *Blood* 101, 3492–3494 (2003).
- Giansily-Blaizot M, Aguilar-Martinez P, Mazurier C *et al.*: Prenatal diagnosis of severe factor VII deficiency using mutation detection and linkage analysis. *Br. J. Haematol.* 112, 251–252 (2001).
- Manco-Johnson MJ, Abshire TC, Shapiro AD *et al.*: Prophylaxis versus episodic treatment to prevent joint disease in boys with severe hemophilia. *N. Engl. J. Med.* 357(6), 535–544 (2007).

Financial & competing interests disclosure

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

No writing assistance was utilized in the production of this manuscript.

27. Mariani G, Konkle BA, Ingerslev J: Congenital factor VII deficiency: therapy with recombinant activated factor VII – a critical appraisal. *Haemophilia* 12, 19–27 (2006).
28. Rodegheiro F, Kadir RA, Tosi A *et al.*: Relevance of quantitative assessment of bleeding in haemorrhagic disorders. *Haemophilia* 14(Suppl. 3), 68–75 (2008).
29. Asselta R, Tenchini ML, Duga S: Inherited defects of coagulation factor V: the hemorrhagic side. *J. Thromb. Haemost.* 4, 26–34 (2006).
30. Mariani G, Bernardi F: Factor VII deficiency. *Semin. Thromb. Hemost.* 35, 400–406 (2009).
31. Gomez K, Bolton-Maggs P: Factor XI deficiency. *Haemophilia* 14, 1183–1189 (2008).
- **Provides an interesting review of the management of inherited factor XI deficiency during surgery.**
32. Brenner B, Kuperman AA, Watzka M *et al.*: Vitamin K-dependent coagulation factors deficiency. *Semin. Thromb. Hemost.* 35, 439–446 (2009).
33. Saunders RE, Perkins SJ: CoagMDB: a database analysis of missense mutations within four conserved domains in five vitamin K-dependent coagulation serine proteases using a text-mining tool. *Hum. Mutat.* 29(3), 333–344 (2008).

Websites

101. Rare Bleeding Disorder Database
www.rbdd.org
102. F XI Deficiency Mutation Database®
www.biochem.ucl.ac.uk/~becky/FXI/links.php
103. Factor XIII Registry Database
www.f13-database.de

Affiliations

- Muriel Giansily-Blaizot
Laboratoire d'hématologie,
CHU de Montpellier, 80 avenue Augustin
Fliche, 34095 Montpellier Cedex 5, France
Tel.: +33 467 337 033
Fax: +33 467 337 036
m-giansily@chu-montpellier.fr
- Jean-François Schved
Laboratoire d'hématologie,
CHU de Montpellier, 80 avenue Augustin
Fliche, 34095 Montpellier Cedex 5, France
Tel.: +33 467 337 033
Fax: +33 467 337 036
jf-schved@chu-montpellier.fr